Section 1: Multiple Choice Questions (MCQs)

1. AWS Core Services

1. C
2. A
3. C

2. IAM & Security

4. A
5. B

3. Networking and Connectivity

6. B
7. A

4. Storage & Databases

8. B
9. B

5. AWS Billing & Pricing

10. C
11. A

6. Research-based AWS Questions - using google only

12. What are AWS Landing Zones, and how do they help with multi-account governance?

    A landing zone is a well-architected, multi-account AWS environment that is scalable and secure. This is a starting point from which your organization can quickly launch and deploy workloads and applications with confidence in your security and infrastructure environment.

13. Explain how AWS WAF protects web applications from common attacks.

    You can use AWS WAF to inspect web requests for matches to conditions that you specify, such as the IP address that the requests originate from, the value of a specific request component, or the rate at which requests are being sent.

    For additional protection against distributed denial of service (DDoS) attacks, AWS also offers AWS Shield Advanced. AWS Shield Advanced provides expanded DDoS attack protection for your Amazon CloudFront distributions, Amazon Route 53 hosted zones, and Elastic Load Balancing load balancers.

14. What is AWS Snowball, and when should it be used?

    Snowball is a petabyte-scale data transport solution that uses secure appliances to transfer large amounts of data into and out of the AWS cloud. Using Snowball addresses common challenges with large-scale data transfers including high network costs, long transfer times, and security concerns.

15. What are the key differences between AWS Backup and manual snapshot backups?

    An AWS snapshot is just a point-in-time copy of an Amazon EBS volume for an EC2 instance with limited storage and recovery options.

    An AWS EC2 backup is a more comprehensive and flexible copy of your cloud workloads, offering reliable protection and ensuring fast and consistent recovery.

16. How does AWS Shield help mitigate DDoS attacks?

    AWS Shield helps protect against DDoS attacks by automatically detecting and mitigating them. It integrates with AWS edge services like CloudFront and Route 53 to reduce response times, uses SYN cookies for protection against SYN Floods, and applies techniques like rate limiting to minimize the impact of attacks.

17. Explain the differences between AWS Transit Gateway and VPC Peering.

    VPC Peering is complex at scale while Transit Gateways simplify network management architecture, reduce operational overhead, and centrally manage external connectivity at scale.

18. What is AWS Step Functions, and how does it help with workflow automation?

    AWS Step Functions makes it easy to coordinate the components of distributed applications as a series of steps in a visual workflow. You can quickly build and run state machines to execute the steps of your application in a reliable and scalable fashion.

19. How does AWS Control Tower assist organizations in managing multiple AWS accounts?

    AWS Control Tower offers a straightforward way to set up and govern an AWS multi-account environment, following prescriptive best practices.

    It uses other services such as Organizations, Service Catalog, and Config to govern the multi-account environment.

20. What is the significance of AWS Outposts in hybrid cloud solutions?

Organizations can use the hardware infrastructure, APIs, tools and management controls available in the AWS cloud on-prem, ensuring a consistent hybrid experience for developers and IT operations. Users can monitor capacity within Outposts to assess system health and performance.

21. Explain the key use cases for AWS Elastic File System (EFS) compared to S3 and EBS.

    EBS used to be accessible to a single EC2 instance only, making it most like your physical hard drive.

    EBS is a block storage service, which means all data within EBS is stored in equally sized blocks.

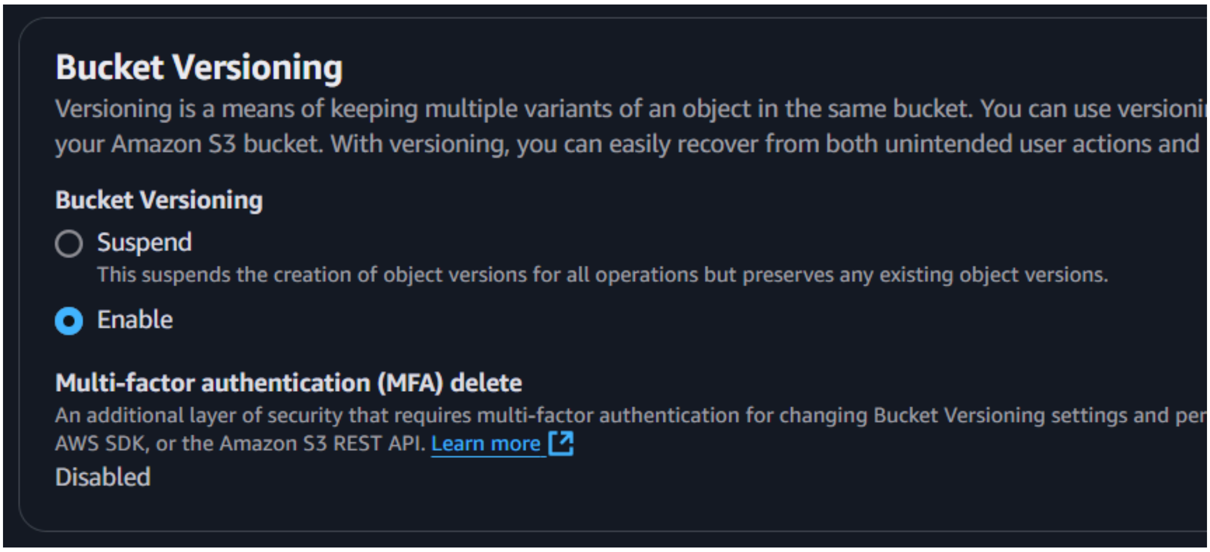
    Unlike EBS, EFS can be mounted by multiple EC2 instances, meaning many virtual machines may store files within an EFS instance.

    S3 is scalable, like EFS, and has access to multiple EC2 instances. However, it can also be accessed by other cloud services, and its object storage system makes it ideal for handling large volumes of static data as well as complex queries.

Section 2: Hands-on UI-Based Questions

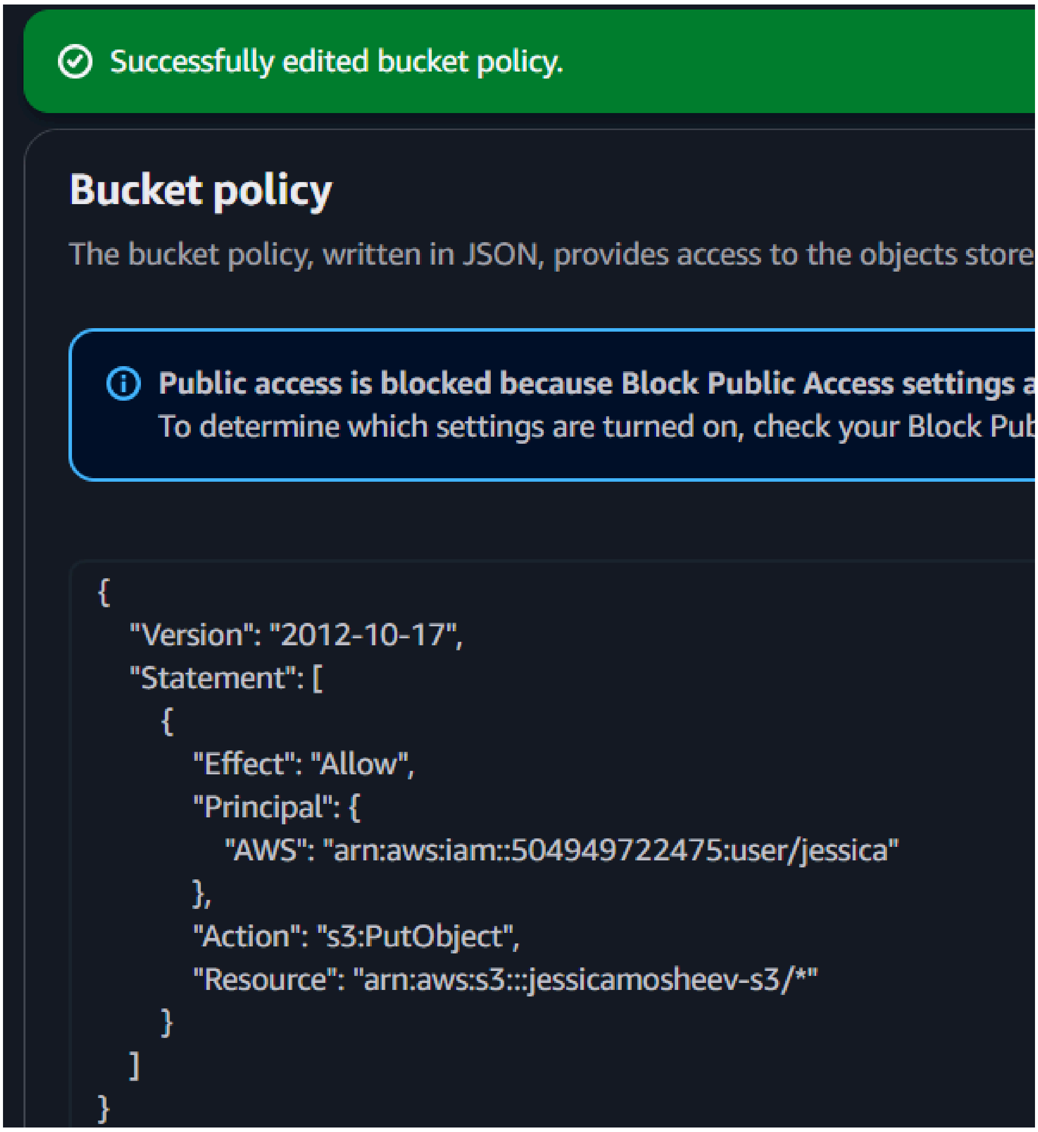
1. S3 Bucket Configuration

- Navigate to the AWS Management Console and create a new S3 bucket.
- Enable versioning and explain the steps you took.



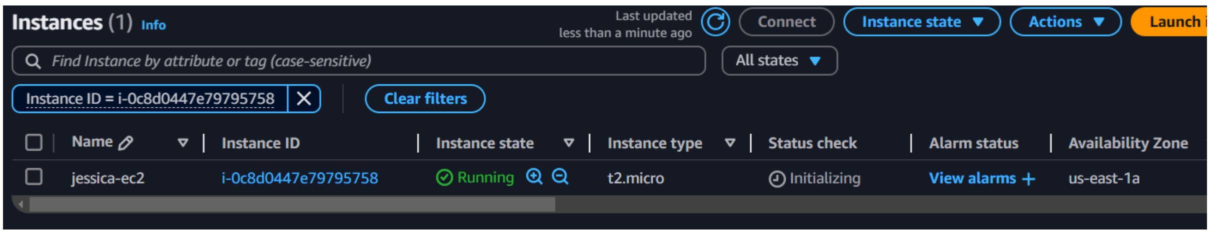
- Set a policy to allow only your IAM user to upload objects.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::504949722475:user/jessica"
      },
      "Action": "s3:PutObject",
      "Resource": "arn:aws:s3:::jessicamosheev-s3/*"
    }
  ]
}
```

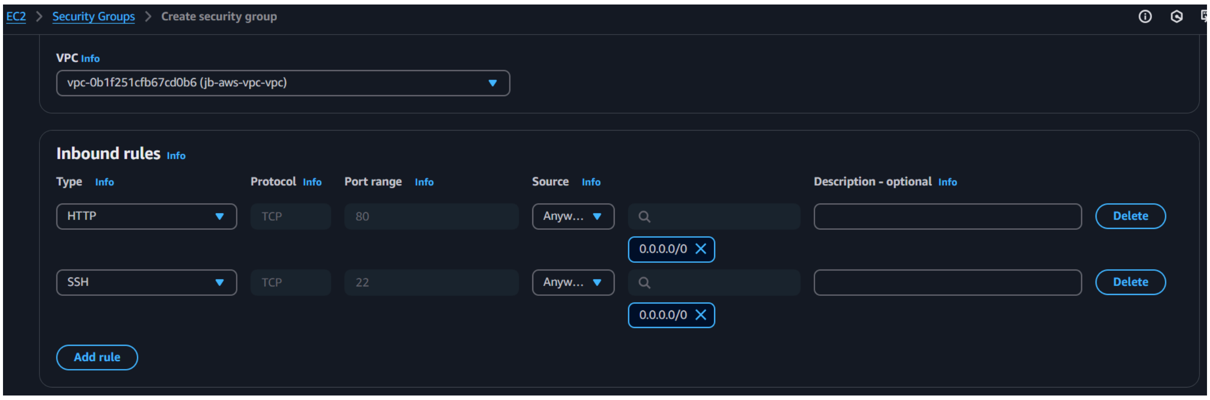


2. Launch an EC2 Instance

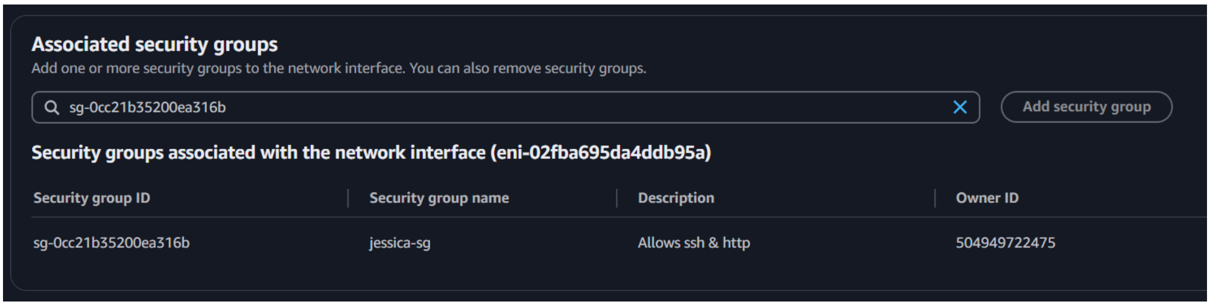
- Using the AWS UI, create a **t2.micro** EC2 instance.



- Attach a Security Group that allows inbound SSH (port 22) and HTTP (port 80) traffic.



נשייך את הsg החדש למכונה שלי ונמחק את הדיפולטיבי



3. Configure an IAM User with S3 Access

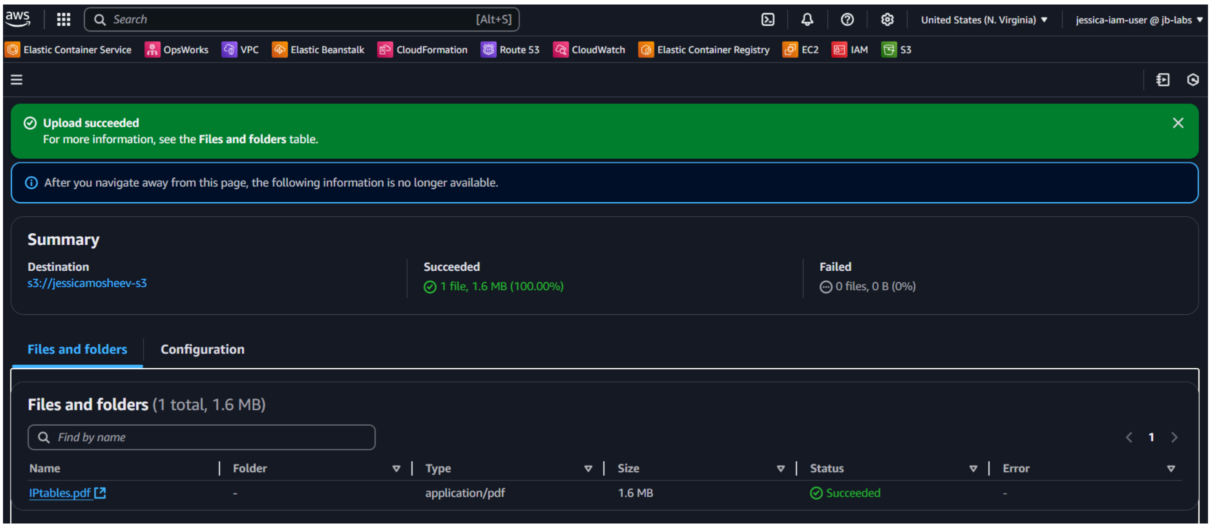
- Create a new IAM user with permissions to access only a specific S3 bucket.

```
{
        "Version": "2012-10-17",
        "Statement": [
                {
                        "Sid": "VisualEditor0",
                        "Effect": "Allow",
                        "Action": [
                                "s3:ListBucket",
                                "s3:GetBucketLocation"
                        ],
                        "Resource": "arn:aws:s3:::*"
                },
                {
                        "Sid": "VisualEditor1",
                        "Effect": "Allow",
                        "Action": [
                                "s3:PutObject",
                                "s3:GetObject",
                                "s3:DeleteObject"
                        ],
                        "Resource": [
                                "arn:aws:s3:::jessicamosheev-s3/*",
                                "arn:aws:s3:::jessicamosheev-s3"
                        ]
```

```
        },
        {
            "Sid": "VisualEditor2",
            "Effect": "Allow",
            "Action": [
                "s3:ListAccessPointsForObjectLambda",
                "s3:ListBucketMultipartUploads",
                "s3:ListAccessPoints",
                "s3:ListCallerAccessGrants",
                "s3:ListBucketVersions",
                "s3:ListJobs",
                "s3:ListBucket",
                "s3:ListMultiRegionAccessPoints",
                "s3:ListStorageLensGroups",
                "s3:ListAccessGrantsLocations",
                "s3:ListMultipartUploadParts",
                "s3:ListStorageLensConfigurations",
                "s3:ListTagsForResource",
                "s3:ListAllMyBuckets",
                "s3:ListAccessGrantsInstances",
                "s3:ListAccessGrants"
            ],
            "Resource": "arn:aws:s3:::*"
        }
    ]
}
```

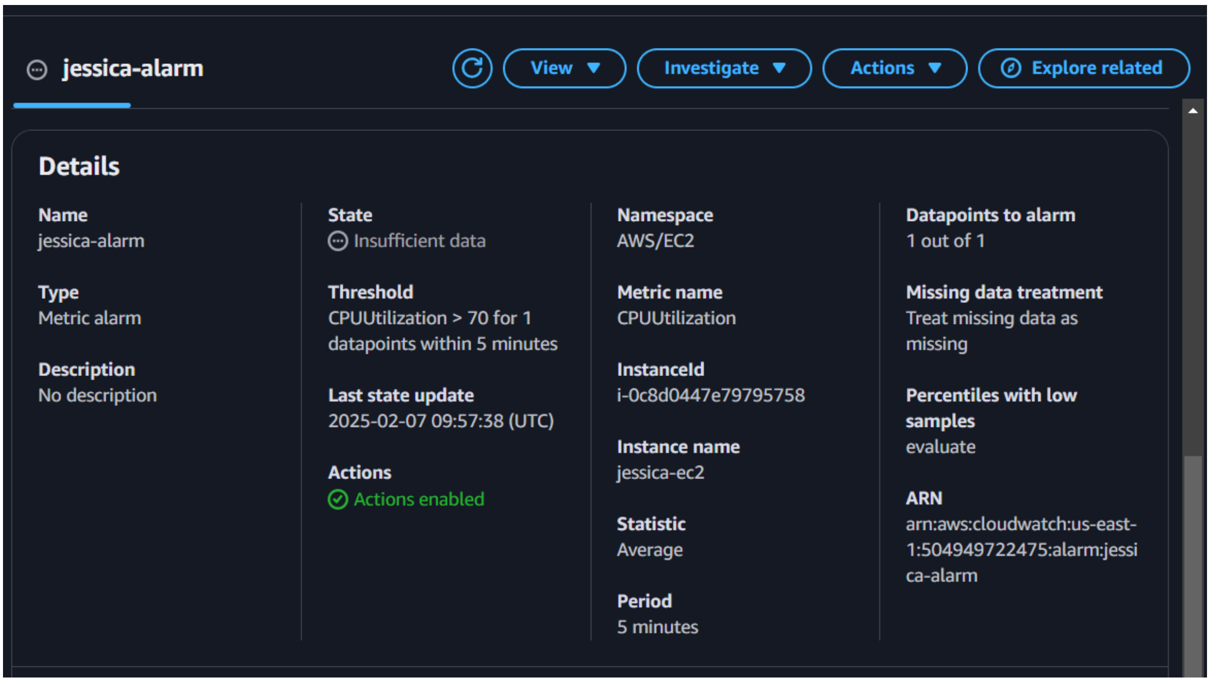
- How would you verify that the user has the correct permissions?

אני אתחבר עם היוזר IAM שיצרתי ואנסה להוסיף קבצים לבאקט שאליו נתתי את ההרשאות



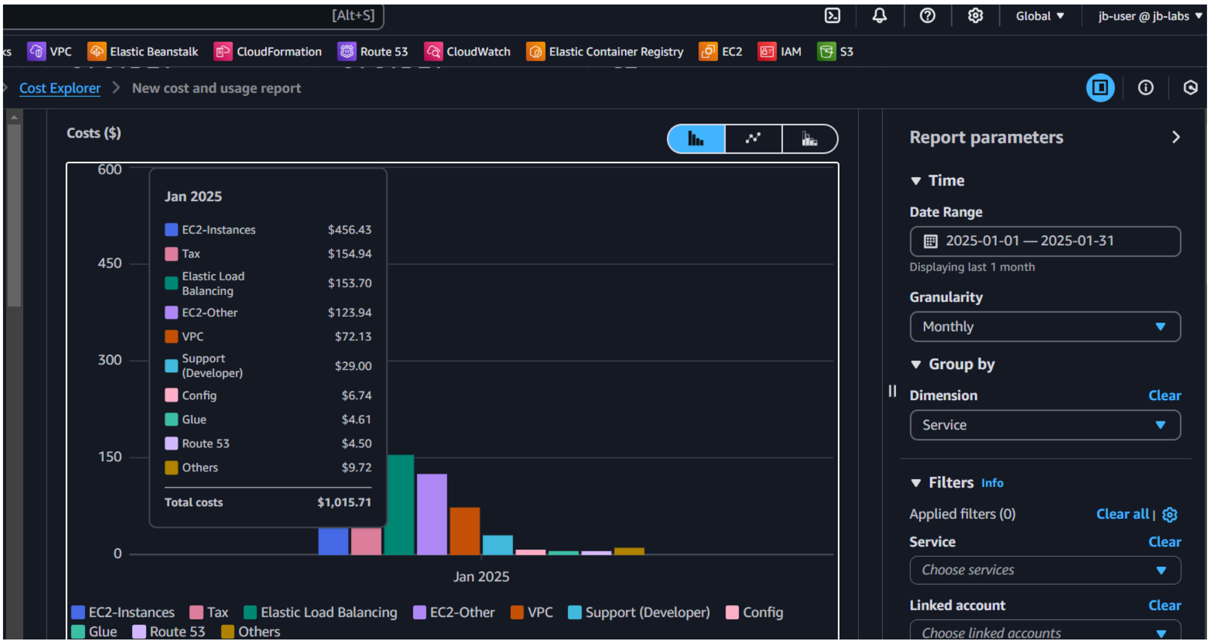
4. Set Up a CloudWatch Alarm

- Navigate to AWS CloudWatch and create an alarm to monitor CPU usage on an EC2 instance.
- Set the alarm to trigger if CPU utilization exceeds **70%** for **5 minutes**.
- Describe how you configured notifications for the alarm.



5. Identify AWS Billing Costs

- Navigate to AWS Cost Explorer and check the billing details for the past month.



- Explain the steps to analyze usage and forecast costs.

To check your billing in AWS Cost Explorer, log in to the AWS Management Console, select "Last Month" as the date range, and set the granularity to Monthly. Group the data by "Service" to analyze costs per service and apply filters as needed for a more detailed view. Use the forecast tool to estimate future expenses based on current usage trends. This helps you identify high-cost services and optimize your spending effectively.

Section 3: Hands-on advanced

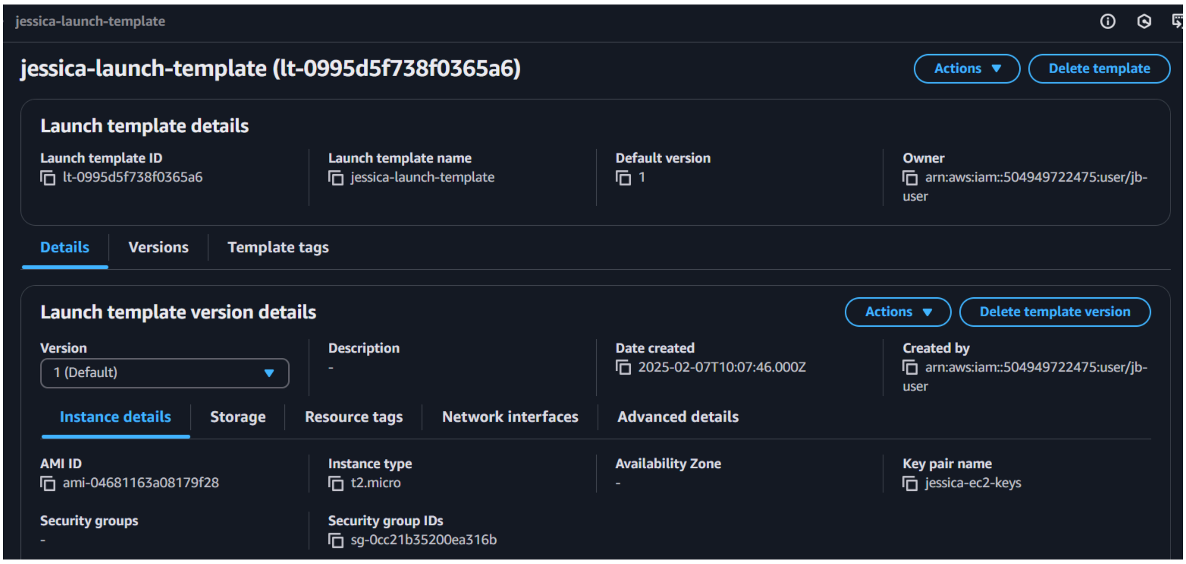
Instructions:

- **Perform the following tasks using the AWS UI.**
- **Provide screenshots as proof of completion.**
- **No AWS CLI is required; all actions must be performed using the AWS Management Console.**

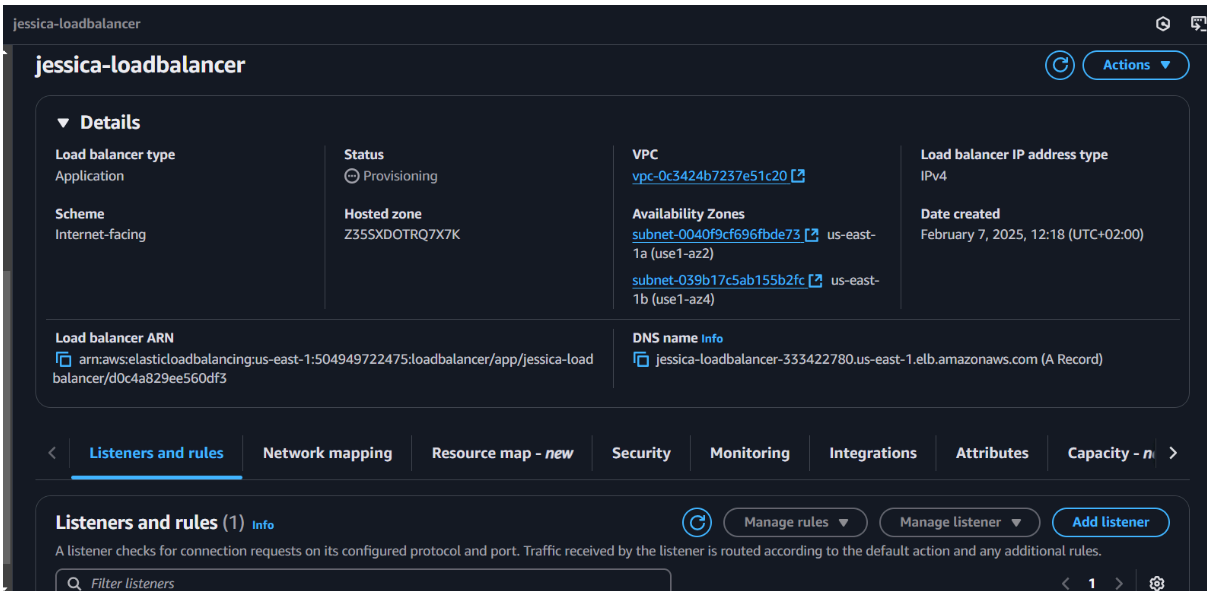
---

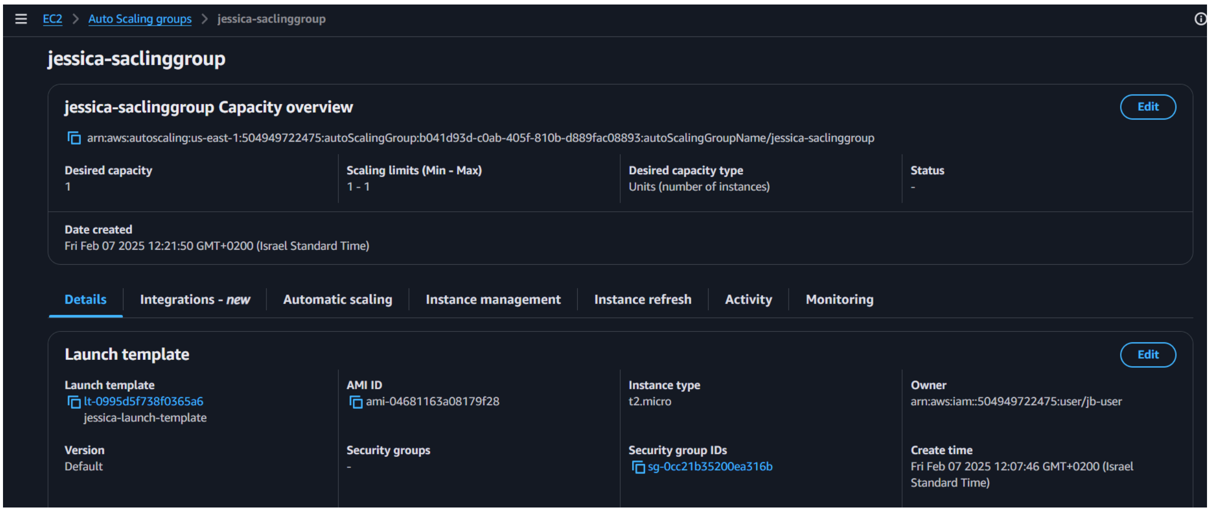
1. Deploy an Auto Scaling Group with a Single EC2 Instance

- Create an Auto Scaling Group with a t2.micro Amazon Linux 2 instance.
- Configure the Launch Template to include the following:
  - Amazon Linux 2 AMI.
  - Security Group that allows inbound SSH (port 22) and HTTP (port 80).



- Set the desired capacity to 1 instance.
- Attach a Load Balancer to the Auto Scaling Group.





2. Connect to the EC2 Instance and Install Nginx

- Connect to the running EC2 instance via SSH.

```
ubuntu@DESKTOP-A44MS8P:~$ chmod 400 jessica-ec2-keys.pem
ubuntu@DESKTOP-A44MS8P:~$ ls -l
total 12
drwxr-xr-x 3 ubuntu ubuntu 4096 Feb  1 20:53 devops-git-lab
-r-------- 1 ubuntu ubuntu 1678 Feb  7 12:43 jessica-ec2-keys.pem
-r-------- 1 ubuntu ubuntu 1674 Jan 31 15:07 newkeypair.pem
ubuntu@DESKTOP-A44MS8P:~$ ssh -i "jessica-ec2-keys.pem" ec2-user@ec2-3-89-140
-142.compute-1.amazonaws.com
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-84-147 ~]$
```

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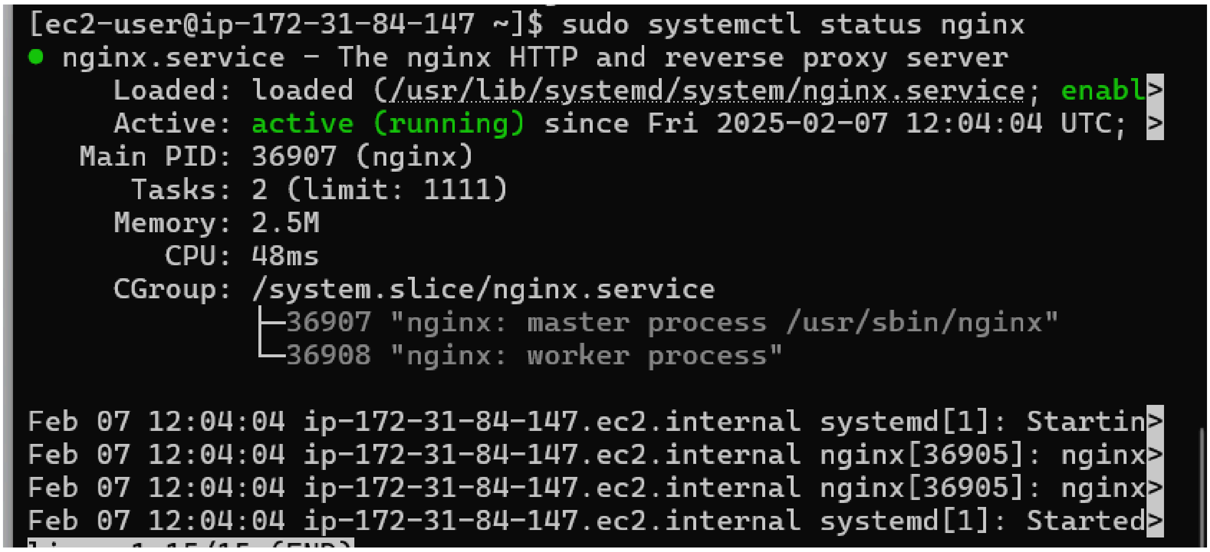
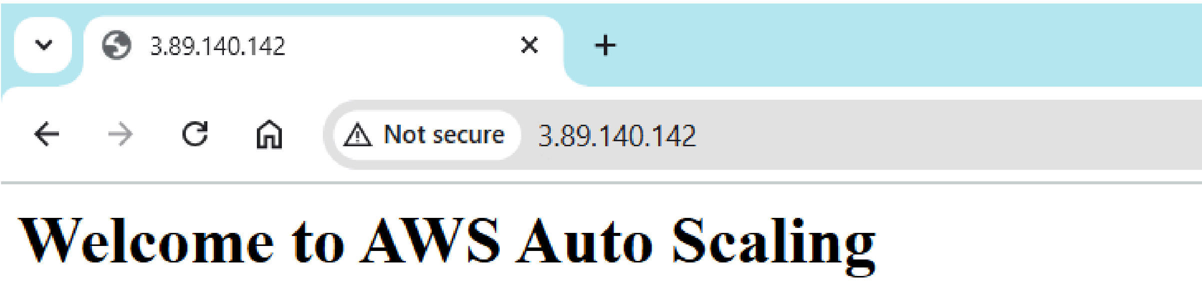
```
ubuntu@DESKTOP-A44MS8P:~$ telnet 3.89.140.142 22
Trying 3.89.140.142...
Connected to 3.89.140.142.
Escape character is '^]'.
SSH-2.0-OpenSSH_8.7
```

Install Nginx and create a simple HTML welcome page - see commands to run below:

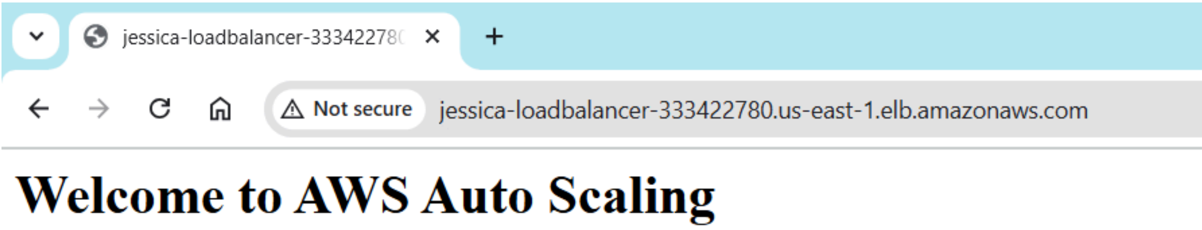

[ec2-user@ip-172-31-84-147 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabl>
Active: active (running) since Fri 2025-02-07 12:04:04 UTC; >
Main PID: 36907 (nginx)
Tasks: 2 (limit: 1111)
Memory: 2.5M
CPU: 48ms
CGroup: /system.slice/nginx.service
├─36907 "nginx: master process /usr/sbin/nginx"
└─36908 "nginx: worker process"
Feb 07 12:04:04 ip-172-31-84-147.ec2.internal systemd[1]: Startin>
Feb 07 12:04:04 ip-172-31-84-147.ec2.internal nginx[36905]: nginx>
Feb 07 12:04:04 ip-172-31-84-147.ec2.internal nginx[36905]: nginx>
Feb 07 12:04:04 ip-172-31-84-147.ec2.internal systemd[1]: Started>


[ec2-user@ip-172-31-84-147 ~]$ curl http://localhost:80
<h1>Welcome to AWS Auto Scaling</h1>


3.89.140.142
Not secure
3.89.140.142
Welcome to AWS Auto Scaling

3. Access the Web Page via the Load Balancer

- Retrieve the Load Balancer DNS name.
- Open a web browser and access the Load Balancer URL.



4. IAM User Setup for S3 Access

- Create a new IAM user with permissions limited to accessing a specific S3 bucket.

```
{
        "Version": "2012-10-17",
        "Statement": [
                {
                        "Sid": "VisualEditor0",
                        "Effect": "Allow",
                        "Action": [
                                "s3:ListBucket",
                                "s3:GetBucketLocation"
                        ],
                        "Resource": "arn:aws:s3:::*"
                },
                {
                        "Sid": "VisualEditor1",
                        "Effect": "Allow",
```

```
                "Action": [
                    "s3:PutObject",
                    "s3:GetObject",
                    "s3:DeleteObject"
                ],
                "Resource": [
                    "arn:aws:s3:::jessicamosheev-s3/*",
                    "arn:aws:s3:::jessicamosheev-s3"
                ]
            },
            {
                "Sid": "VisualEditor2",
                "Effect": "Allow",
                "Action": [
                    "s3:ListAccessPointsForObjectLambda",
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                    "s3:ListAccessPoints",
                    "s3:ListCallerAccessGrants",
                    "s3:ListBucketVersions",
                    "s3:ListJobs",
                    "s3:ListBucket",
                    "s3:ListMultiRegionAccessPoints",
                    "s3:ListStorageLensGroups",
                    "s3:ListAccessGrantsLocations",
                    "s3:ListMultipartUploadParts",
                    "s3:ListStorageLensConfigurations",
```
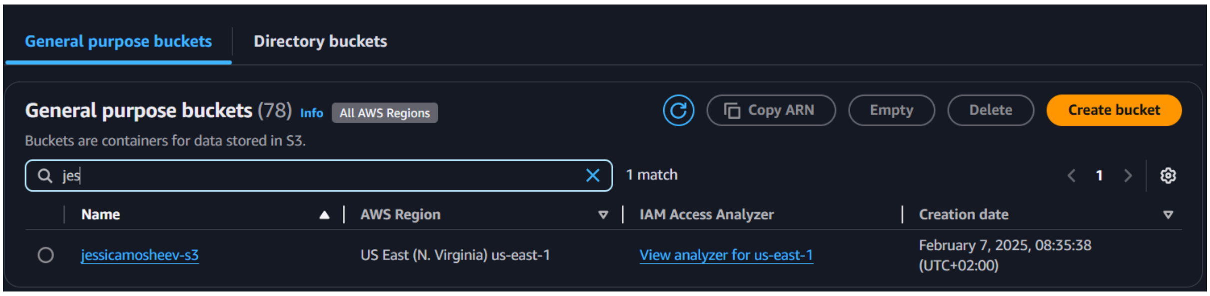
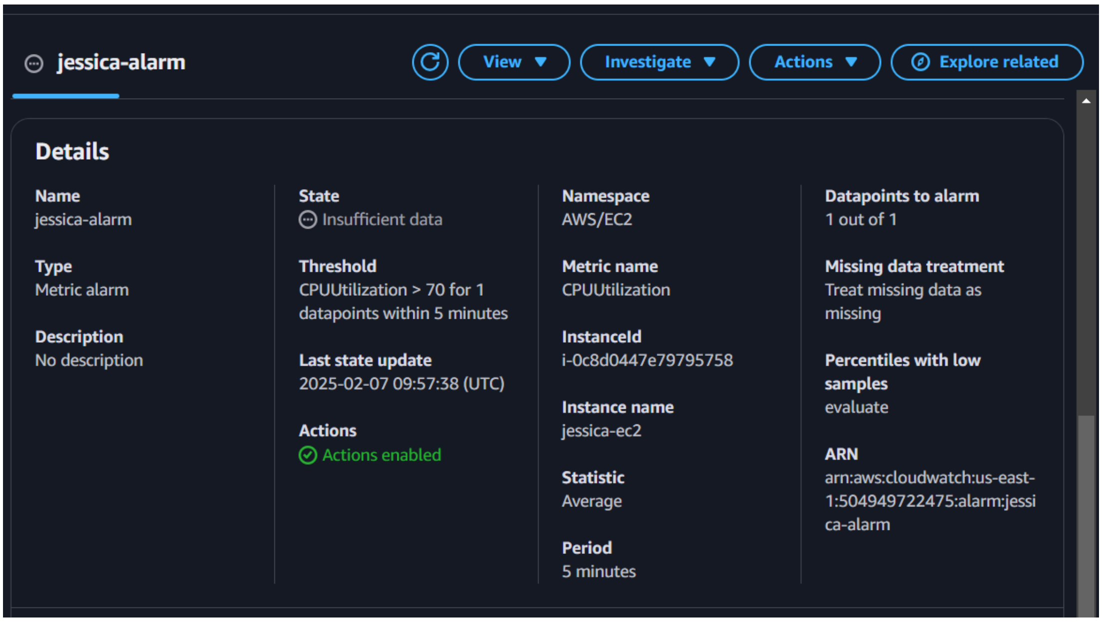

```
                                        "s3:ListTagsForResource",
                                        "s3:ListAllMyBuckets",
                                        "s3:ListAccessGrantsInstances",
                                        "s3:ListAccessGrants"
                              ],
                              "Resource": "arn:aws:s3:::*"
                    }
          ]
}
```

- Log in as the IAM user and verify access to the S3 bucket.
- 

5. Create a CloudWatch Alarm for CPU Usage

- Navigate to AWS CloudWatch and create an alarm for the EC2 instance.
- Set the alarm to trigger when CPU utilization exceeds 70% for 5 minutes.
- Configure notifications via email (SNS).
- Provide a screenshot of the CloudWatch alarm configuration.

A CloudWatch Alarm for CPU Usage is a monitoring tool within AWS CloudWatch that tracks the percentage of CPU utilization for an Amazon EC2 instance or other supported AWS services. It automatically triggers notifications or predefined actions when the CPU usage crosses a specified threshold.


jessica-alarm
View
Investigate
Actions
Explore related
Details
Name
jessica-alarm
Type
Metric alarm
Description
No description
State
Insufficient data
Threshold
CPUUtilization > 70 for 1 datapoints within 5 minutes
Last state update
2025-02-07 09:57:38 (UTC)
Actions
Actions enabled
Namespace
AWS/EC2
Metric name
CPUUtilization
InstanceId
i-0c8d0447e79795758
Instance name
jessica-ec2
Statistic
Average
Period
5 minutes
Datapoints to alarm
1 out of 1
Missing data treatment
Treat missing data as missing
Percentiles with low samples
evaluate
ARN
arn:aws:cloudwatch:us-east-1:504949722475:alarm:jessica-alarm